6.7.2 Angle of minimum deviation

A graph plotted between the angle of incidence and angle of deviation is shown in Figure 6.41. One could observe that the angle of deviation decreases with increase in angle of incidence and reaches a minimum value D and then continues to increase.

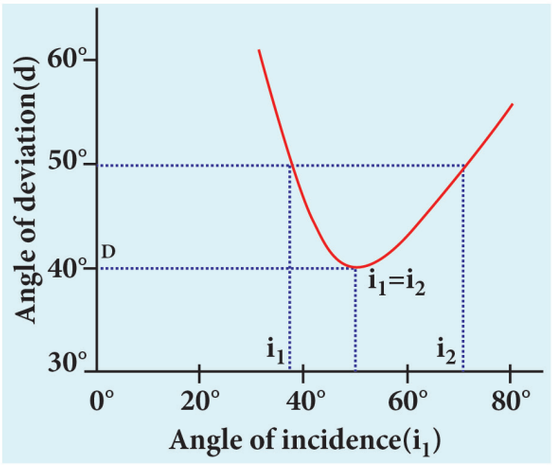


Figure 6.41 Graph between i and d

The minimum value of angle of deviation is called angle of *minimum deviation D*. At minimum deviation,

(i) the angle of incidence is equal to the angle of emergence, $i_1 = i_2$.

(ii) the angle of refraction at the face one and face two are equal, $r_1 = r_2$).

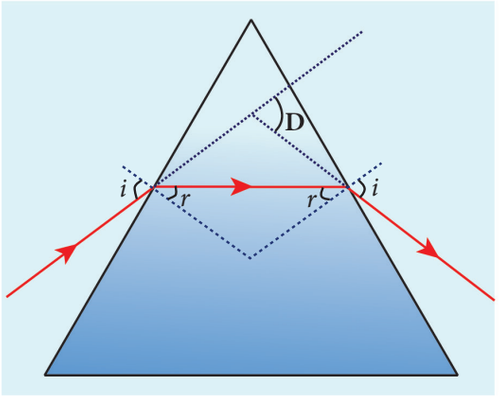


Figure 6.42 Angle of minimum deviation

(iii) the refracted ray inside the prism is parallel to its base of the prism.

The case of angle of minimum deviation is shown in Figure 6.42.

6.7.3 Refractive index of the material of the prism

At minimum deviation, $i_1 = i_2 = i$ and $r_1 = r_2 = r$

Now, the equation (6.88) becomes,

$$D = i_1 + i_2 - A \Rightarrow 2i - A \quad \text{(or)} \quad i = \frac{(A+D)}{2}$$

The equation (6.87) becomes,

$$r_1 + r_2 = A \Rightarrow 2r = A \quad \text{(or)} \quad r = \frac{A}{2}$$

Substituting i and r in Snell's law, $n = \frac{\sin i}{\sin r}$

$$n = \frac{\sin\left(\frac{A+D}{2}\right)}{\sin\left(\frac{A}{2}\right)} \qquad (6.89)$$

The above equation is used to determine the refractive index of the material of the prism. The angles A and D can be measured experimentally.

EXAMPLE 6.21

The angle of minimum deviation for an equilateral prism is 37°. Find the refractive index of the material of the prism.

Solution

Given, $A = 60°$; $D = 37°$

Equation for refractive index is,

$$n = \frac{\sin\left(\frac{A+D}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Substituting the values,

XII_Physics_Vol-2_UNIT-6_EM.indd 40 26-12-2022 11:38:37